Maximizing Areal-Density: a frequency domain perspective

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This paper takes a frequency domain perspective on the task of choosing the track-density and bit density for a hard disk drive (HDD). A simple metric is derived that requires only that the wanted signal spectrum be cleanly separated - because it is either repetitive or is based on a known data pattern. The metric, referred to as 'spectral capacity', is an approximation based on Shannon's famous formula, log(1+SNR), together with the constraint of binary signaling. The metric is compared with mutual-information - which is a more direct measure but one that requires an actual data-detector to be in place. The comparison is made both with waveforms from a simple recording simulation and with waveforms from an HDD employing heat-assisted magnetic recording (HAMR). In general, 'spectral-capacity' is found to track the detector-based metric closely over a wide range of TPI, BPI, noise mix and even a simple modulation code.

***Index Terms*—Hard Disk Drives, Heat Assisted Magnetic Recording, Recording Channel, Shannon Capacity**

I. Introduction

Hard Disk Drives (HDD) can contain as many as 20 separate head and disk-surface combinations, each with its own behavior as a function of radius (zone). In setting the final customer capacity, it is important to gauge the areal-density capability of each head/disk/zone and appropriately share the areal-density burden across all the surfaces. The areal density capability of a given head/disk/zone is a complex function of many parameters most of which are 'fixed' in the components, but many of which are determined during the manufacturing process and some also may be adapted in the customer's office.

Generally, the manufacturing process will test a series of track-densities (TPI) and bit-densities (BPI) as well as other critical parameters such as write-current and laser-current. We are looking for whatever combination delivers the highest user areal-density (TPI x BPI) to later determine how many GBytes we can safely pack into a given head/disk/zone.

The magnetic recording channel is far from ideal. There are typically signal-nonlinearities, interfering signals from adjacent tracks, signal-dependent media noise, and head/electronic noise. All this is much clearer when we consider the waveforms in frequency domain. In particular, we derive an approximation for the potential information density based on the narrow-band signal-to-noise ratio vs. frequency. The approximation is based on Shannon's famous expression, log(1+S/N), modified by the constraint of binary recording. This analysis does not depend on the presence of a data detector, but only on the fact that the wanted signal can be cleanly separated, either because it is repetitive or because it arises from a known written pattern.

We refer to this approximation as 'spectral capacity' and use it to make comparisons with measures such as mutual information from a software data-detector. This study is conducted both on waveforms from a simple recording simulation and also on waveforms taken up to ultra-high densities on an HDD employing heat-assisted magnetic recording (HAMR). The latter gives interesting results even up to track-densities where the reader is spanning several data-tracks. In general, spectral capacity is found to track traditional detector-based metrics closely over a wide range of TPI and BPI and also over codes that provide for reduced transition density. It even behaves reasonably with high levels of reader distortion.

II. "Spectral Capacity"

Shannon's formula is $C = B.\log(1 + S/N)$, where C is capacity, B is the available bandwidth, S is the signal power and N is the noise power [1]. This formula applies to a linear analog channel with additive white Gaussian noise (AWGN). As such, it can be expanded into an integral form:

$$C = \int_0^\infty log_2\,(1 + SNR(f))\,df \qquad (1)$$

where, if f is expressed as a fraction of the clock-rate, then the capacity is in information bits per channel-use or per clock-cycle. This capacity might be a number like, for example, 0.85, suggesting that 15% coding redundancy might be required to ensure reliable customer data.

In regions of very low SNR(f) << 1, capacity increases linearly with added signal. But in regions of high SNR >>1, the capacity increases logarithmically and the law of diminishing returns applies. As a consequence, it is better to spread the available power broadly across the useful bandwidth rather than concentrating it in a narrow portion of the spectrum.

Inevitably there are other limitations too. There is always a constraint on power and on how that power is distributed over the band, and, in the case of magnetic recording, only binary signals are ever considered ('non-saturation' recording inevitably results in much higher noise levels).

The Shannon capacity calculation using (1) can easily exceed 1 bit per clock-cycle, which a binary channel can obviously not support. To include the binary signaling limitation we first calculate the Shannon capacity using the integral expression (1) and then find the equivalent single SNR number for a broadband AWGN channel by inverting the expression, C = ½log2(1+SNR) to give SNR = 2^(2C) – 1. We then take this SNR number and interpret it as the SNR for asymmetric binary signaling on an AWGN channel and thence derive a capacity.

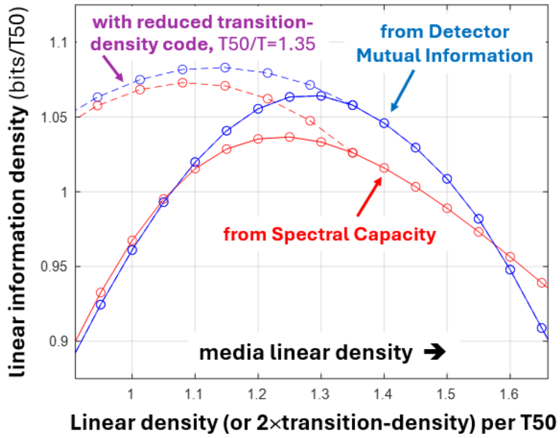

Fig. 1. 'Information Densities' based on Spectral Capacity and detector Mutual Information compared on a 1D recording simulation. Horizontal axis is 2x transition density either for random NRZ or for the reduced transition density code and vertical axis is a user information density.

There is no simple closed-form expression of the capacity for binary signaling over a wide-band AWGN, but the integral expression is readily calculated or it can be closely approximated with a suitable fitting function.

$$C_B = 1 - \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(z-\sqrt{SNR_b})^2}{2}} log_2(1 + e^{-2z\sqrt{SNR_b}})dz \quad (2)$$

We call the resulting metric a 'capacity' though it refers to a specific SNR spectrum rather than some maximum over all possibilities for choice of written signal. More strictly, it is an approximation for the mutual information between the known/wanted signal and the noisy readback waveform. It can be compared directly with the mutual information (MI) calculated between the written binary data and the soft decision information (log-likelihood ratios) at the output of a conventional data detector [4].

III. Results and Discussion

The effectiveness of this approximation is first explored using a 1D Monte-Carlo simulation that uses error-function transitions and additive media noise (divided into stationary and jitter-noise components) and additive white noise to represent head-electronics noise. The noise powers were fixed in the ratio 20%/70%/10%, respectively and the SNR was fixed at 11.5 dB (base-peak). Fig. 1 compares information density derived from spectral capacity and from detector mutual information as linear density is scanned from T50/T = 0.9 to 1.7. The curves are similar, reaching maxima that differ in magnitude by about 3% and in position by about 4%. The dashed curves show the advantage of a reduced transition-density code from T50/T = 1.35. In all cases, the x-axis is 2×average transitions per T50.

For further testing, waveforms were gathered from a contemporary HAMR HDD over a very wide range of track-density and bit-density as detailed in [2,3]. Fig. 2 compares information density derived from (a) spectral capacity and (b) detector mutual information as a function of track-density and linear density. Again, the general shape and the peak magnitude and position are quite similar. One general observation in both figures is that the detector rolls off quite rapidly at high linear density (T50/T). It is likely that the 16-state SOVA detector becomes suboptimal at this very high level of dispersion.

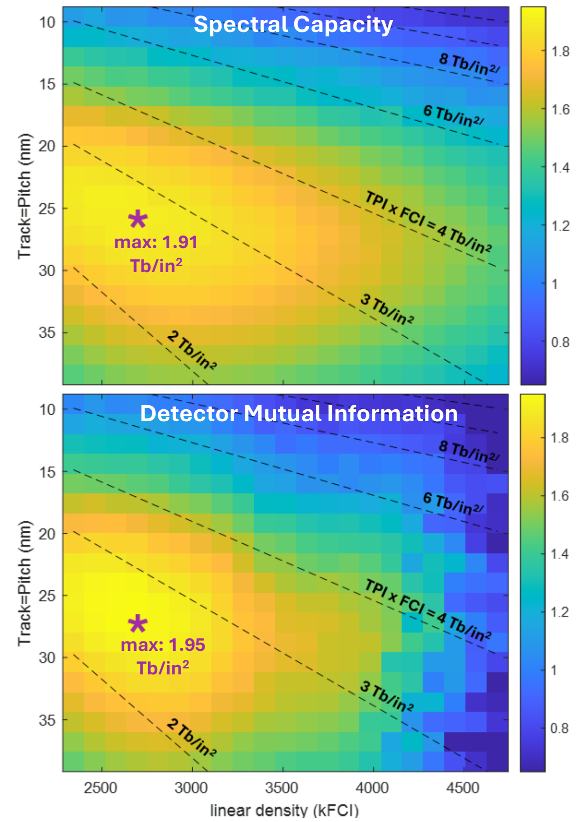

Fig. 2. Information densities based on Spectral Capacity (upper) and Detector Mutual Information (lower) are compared on a contemporary HAMR HDD as a function of track-density and linear density (kFCI). The maximum occurs in each case at very close to the same TPI and BPI. (The colorbar is user information density in Tbits /sq.in.)

In conclusion, we have developed a novel metric that estimates the information capacity of a magnetic recording channel. The metric relies on identifying the wanted signal within the recovered waveform but does not require an actual detector to be implemented. The metric was tested both on a software channel simulation and on waveforms gathered on a contemporary HAMR HDD. The metric provides surprisingly good guidance towards setting the operating point (TPI, BPI) for the tested HDD. This is despite the approximation making no allowance for the non-Gaussian, signal-dependent nature of the noise. The results seem to suggest that the performance of a recording channel is primarily set by its overall spectral response as much as by minimum distance events and the occurrence of multiple consecutive transitions.

References

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